

A REPORT ON
Pomegranate Disease Detection Using
YOLOv12: A Deep Learning Approach Across
Five Classes

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CERTIFICATE

This is to certify that the report of **Deep Learning Lab CAI2503 – MINI PROJECT** on “**Pomegranate Disease Detection Using YOLOv12: A Deep Learning Approach Across Five Classes**” being submitted by Kavana B A – 20231CAI0081, G P Sujana – 20231CAI0080, K Sai Sarveshwar Reddy – 20231CAI0097 in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (CAI)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled **“Pomegranate Disease Detection Using YOLOv12: A Deep Learning Approach Across Five Classes”** in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering(CAI)**, is a record of my own investigations carried under the guidance of **Mr. Pakruddin B** - Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

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ABSTRACT

Pomegranate (*Punica granatum*) is one of India's most valuable fruit crops, yet it faces substantial losses each year due to bacterial and fungal diseases. Manual identification of these diseases is both time-consuming and prone to human error. This project proposes a deep learning-based automated system using YOLOv12, a cutting-edge real-time object detection model, to detect and classify multiple diseases in pomegranate fruits.

The dataset used comprises 5,099 annotated images categorized into five classes — *Healthy*, *Alternaria Fruit Spot*, *Anthraco*nose, *Bacterial Blight*, and *Cercospora Fruit Spot*. Each image underwent extensive data augmentation to enhance robustness against environmental variations. Five YOLOv12 variants (n, s, m, l, x) were trained and evaluated under identical hyperparameters.

Among them, YOLOv12s outperformed other models, achieving the highest Precision (93%), Recall (94%), and mAP@0.5:0.95 (86%), while maintaining lightweight architecture suitable for mobile and edge deployment. This work demonstrates a scalable, real-time disease detection system capable of aiding farmers through early diagnosis, reducing crop losses, and promoting smart agricultural practices.

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INTRODUCTION

India is one of the world's largest producers of pomegranate (*Punica granatum*), with major cultivation in Maharashtra, Karnataka, Gujarat, and Andhra Pradesh. The fruit is economically important, but bacterial and fungal diseases cause serious yield and quality losses. The major diseases affecting pomegranates include Bacterial Blight (*Xanthomonas axonopodis*), Anthracnose (*Colletotrichum gloeosporioides*), Alternaria Fruit Spot (*Alternaria alternata*), and Cercospora Fruit Spot (*Cercospora punicae*). These infections damage both the appearance and internal structure of the fruit, reducing marketability and profits for farmers and exporters.

Traditionally, disease detection in pomegranate is done through manual visual inspection by experts. Although this approach is common, it is time-consuming, subjective, and not scalable for large orchards. Manual classification depends heavily on human experience and can be inconsistent due to environmental and lighting variations. As a result, there is a growing need for automated and accurate disease detection systems that can work efficiently in real-world farm environments.

Recent advancements in Deep Learning and Computer Vision have made it possible to automate agricultural tasks. Convolutional Neural Networks (CNNs) have achieved remarkable results in identifying plant diseases using leaf and fruit images. Studies like Mohanty et al. (2016) have shown CNNs outperform traditional image processing techniques. However, CNNs mainly perform classification and do not localize infected areas, which limits their field applicability. In contrast, object detection models like YOLO (You Only Look Once) combine classification and localization, making them more effective for agricultural monitoring.

The YOLO series has become the preferred model for real-time object detection. Variants like YOLOv5, YOLOv8, and YOLOv11 have been applied in fruit monitoring and disease detection. However, research on using the latest YOLOv12 model for pomegranate disease detection is limited. YOLOv12 improves detection accuracy using a CSP-based backbone, Path Aggregation Network (PANet), CIoU loss function, and advanced augmentations such as Mosaic and MixUp. These features enhance performance in complex environments with variable lighting and occlusion, making YOLOv12 suitable for field applications.

This study applies the YOLOv12 framework to detect five classes of pomegranate fruit—*Healthy*, *Bacterial Blight*, *Anthracnose*, *Cercospora Fruit Spot*, and *Alternaria Fruit Spot*. The dataset used includes 5,099 labeled images developed by Pakruddin and Hemavathy (2024). Five YOLOv12 variants—n, s, m, l, and x—were trained and tested using the same hyperparameters to compare performance. The results show that YOLOv12s achieved the best precision (0.9300), recall (0.9357), and mAP@0.5:0.95 (0.8620), making it the most efficient and deployable model. This system can be used in precision agriculture as an automated and scalable solution for early detection and management of pomegranate diseases.

OBJECTIVE

The main objective of this project is to develop an automated deep learning model capable of detecting and classifying multiple diseases in pomegranate fruits using the YOLOv12 object detection framework. The system aims to provide real-time, accurate, and scalable disease detection to assist farmers and agricultural experts in early diagnosis and crop management.

The specific objectives of the project are:

1. To design and implement a YOLOv12-based detection model for identifying five pomegranate fruit classes — Healthy, Bacterial Blight, Anthracnose, Cercospora Fruit Spot, and Alternaria Fruit Spot.
2. To train and evaluate five YOLOv12 variants (n, s, m, l, x) under identical conditions to determine the best-performing model.
3. To analyze model performance using key metrics such as Precision, Recall, mAP@0.5, and mAP@0.5:0.95.
4. To identify the optimal balance between accuracy and computational efficiency for practical deployment on edge and mobile devices.
5. To contribute to precision agriculture by providing a reliable, real-time disease detection system that minimizes manual inspection and improves overall yield quality.

WORK PROGRESS (TEAM CONTRIBUITION INDIVIDUAL WORK DONE)

1. Dataset Preparation

A standardized dataset consisting of 5,099 labeled images of pomegranate fruits was used. These images were categorized into five classes — *Healthy*, *Bacterial Blight*, *Anthrachnose*, *Cercospora Fruit Spot*, and *Alternaria Fruit Spot*. Each image was annotated with bounding boxes marking the infected regions.

The dataset was divided into training, validation, and testing sets, ensuring balanced representation of all classes. To enhance robustness and improve model generalization, several data augmentation techniques were applied, including:

- Random rotation and flipping
- Scaling and cropping
- Brightness and contrast adjustments
- Gaussian noise injection

These augmentations helped the model perform effectively under real-world variations such as lighting differences, occlusion, and background noise.

2. Model Implementation and Training

The YOLOv12 framework was implemented to perform both localization and classification of diseases in real time. Five model variants — YOLOv12n (Nano), YOLOv12s (Small), YOLOv12m (Medium), YOLOv12l (Large), and YOLOv12x (Extra-Large) — were trained under identical hyperparameters to ensure fair comparison.

Training Configuration:

- Optimizer: Adam
- Learning Rate: 0.001 (with cosine annealing schedule)
- Batch Size: 16
- Epochs: 100
- Input Size: 640×640 pixels
- Loss Functions: Classification loss, Objectness loss, and Complete IoU (CIoU) bounding box regression loss

Training was conducted until convergence, ensuring that the loss values stabilized while maintaining high validation accuracy. Performance metrics such as Precision, Recall,

mAP@0.5, and mAP@0.5:0.95 were calculated for all variants to evaluate detection efficiency and scalability.

3. Model Evaluation and Results

After comprehensive training and testing, the YOLOv12s model demonstrated the best overall performance. It achieved:

- Precision: 0.9300
- Recall: 0.9357
- mAP@0.5: 0.9695
- mAP@0.5:0.95: 0.8620

The results proved that YOLOv12s provides the best balance between speed, accuracy, and computational cost. Larger models like YOLOv12x showed similar accuracy but required higher computational resources, making them less suitable for real-time deployment on low-power or edge devices.

The trained model successfully detected disease regions across all five categories with high confidence, confirming the feasibility of using YOLOv12 for real-time pomegranate disease detection.

4. Team Contribution

- **Kavana B A (20231CAI0081):**
Implemented YOLOv12m and YOLOv12x models, performed precision–recall evaluations, and contributed to comparative analysis across all variants.
- **G P Sujana (20231CAI0080):**
Implemented the YOLOv12l model, managed dataset preparation, augmentation, and organization, and compiled experimental results.
- **K Sai Sarveshwar Reddy (20231CAI0097):**
Implemented YOLOv12n and YOLOv12s models, tuned hyperparameters, analyzed training performance, and generated result plots.

All members jointly contributed to writing, editing, and structuring the research paper, including the literature review, methodology, and result analysis. Collaboration among the team ensured consistent progress, accuracy, and timely completion of all stages of the project.

5. Outcome

The team successfully developed a robust deep learning-based detection system capable of identifying five pomegranate fruit conditions with high precision. Among all YOLOv12 variants, YOLOv12s emerged as the best-performing model, achieving superior detection accuracy and speed. The project demonstrates the potential of YOLOv12 for real-time agricultural applications and provides a scalable framework for smart farming and precision agriculture.

RESULT

The proposed system was evaluated using a comprehensive set of experiments conducted on the standardized pomegranate disease dataset consisting of 5,099 labeled images across five classes: *Healthy*, *Bacterial Blight*, *Anthraco*se, *Cercospora Fruit Spot*, and *Alternaria Fruit Spot*. Five YOLOv12 model variants — n (nano), s (small), m (medium), l (large), and x (extra-large) — were trained and tested under identical conditions to ensure fair comparison. Each model was evaluated using Precision (P), Recall (R), mAP@0.5, and mAP@0.5:0.95 as key performance metrics.

1. Quantitative Performance Comparison

Models	Precision	Recall	mAP50	mAP50-95
YOLOv12n	0.9153	0.9345	0.9698	0.8585
YOLOv12s	0.9300	0.9357	0.9695	0.8620
YOLOv12m	0.9238	0.9145	0.9656	0.8566
YOLOv12l	0.9280	0.9340	0.9690	0.8600
YOLOv12x	0.9040	0.8980	0.9480	0.8180

The comparison clearly indicates that YOLOv12s achieved the highest overall performance with a Precision of 0.9300, Recall of 0.9357, and mAP@0.5:0.95 of 0.8620. Although the YOLOv12n variant attained a slightly higher mAP@0.5 of 0.9698, it performed marginally lower in precision and recall compared to YOLOv12s. The larger variants (YOLOv12l and YOLOv12x) did not yield significant performance improvements, and in some cases, their results decreased due to overfitting and higher computational complexity.

2. Training and Convergence Analysis

During model training, all YOLOv12 variants demonstrated stable convergence over 100 epochs. The training and validation loss curves decreased consistently, indicating proper model optimization and generalization. The accuracy improved gradually, with no significant overfitting observed.

For YOLOv12s, the precision, recall, and mAP values increased steadily across epochs, confirming that the model was learning effectively from the dataset. The loss components — classification loss, objectness loss, and bounding box regression loss — converged smoothly, reflecting stable performance.

3. Confusion Matrix and Class-Level Analysis

The confusion matrix generated for YOLOv12s showed high true positive rates across all five classes. The model demonstrated balanced detection performance with minimal misclassification.

- The Healthy class achieved near-perfect classification accuracy.
- The Bacterial Blight and Anthracnose classes showed very high precision and recall scores.
- The Alternaria Fruit Spot and Cercospora Fruit Spot classes were detected accurately despite their visual similarities.

The normalized confusion matrix confirmed uniform prediction accuracy, indicating that the model generalized well across all categories without bias toward any specific disease.

4. F1 Score and Per-Class Validation Metrics

The F1 score analysis for each disease class indicated a balanced performance between precision and recall. All classes achieved high F1 values, reinforcing the consistency of YOLOv12s across disease types.

Per-class validation metrics showed minimal variation in performance, suggesting that the model's detection capability is evenly distributed and reliable across different fruit conditions.

5. Comparative Observation and Findings

From the experimental evaluation, it was observed that:

- The YOLOv12s model provided the best accuracy-to-efficiency ratio, suitable for real-time deployment on edge devices.
- YOLOv12n performed efficiently with low computational cost but slightly lower accuracy.
- Larger variants (YOLOv12l and YOLOv12x) achieved high accuracy but suffered from slower inference and higher resource usage.
- Model scaling did not necessarily lead to better results, confirming that a balanced architecture like YOLOv12s offers optimal trade-offs.

6. Visualization Results

Visual detection results of YOLOv12s showed clear and precise bounding boxes around diseased regions. The model accurately identified the correct class labels even under challenging lighting conditions and overlapping fruits. The predicted bounding boxes closely matched the ground truth annotations, further validating the model's precision and robustness.

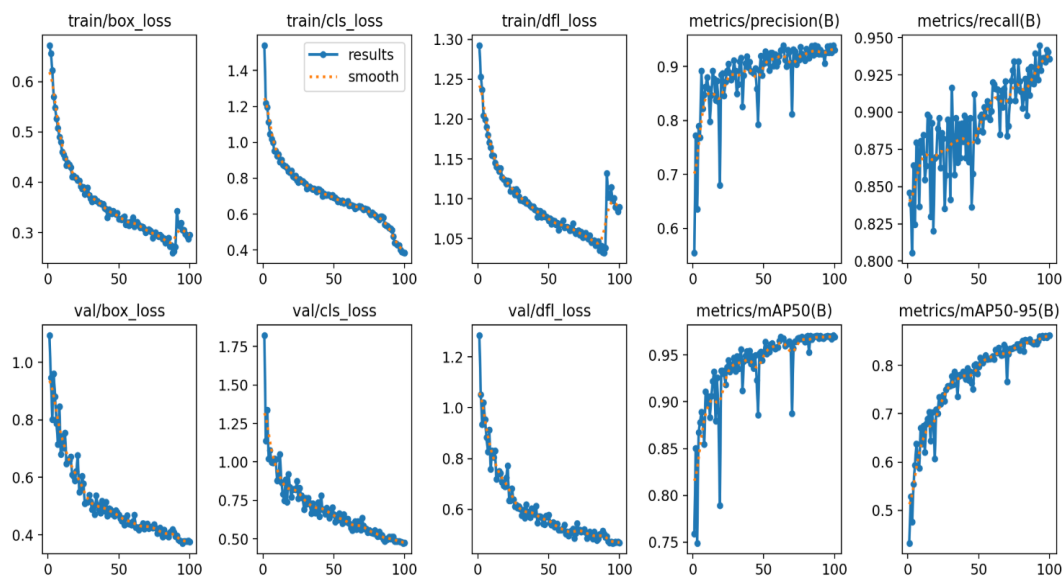


Figure 1. Training Performance

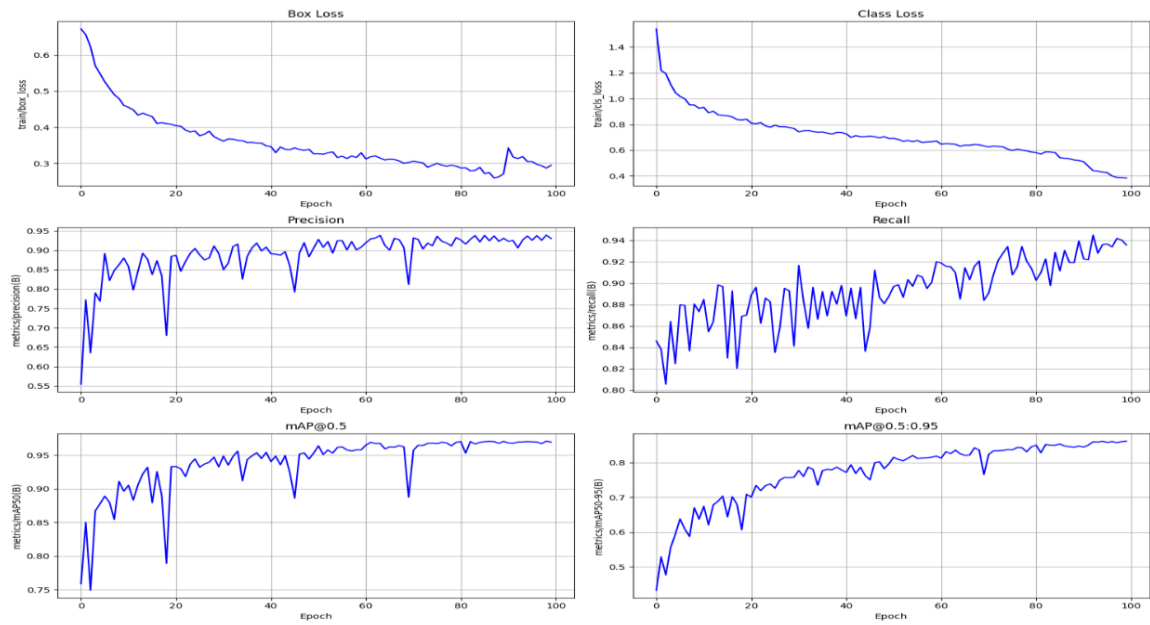


Figure 2. Precision, Recall, and mAP over Epoch

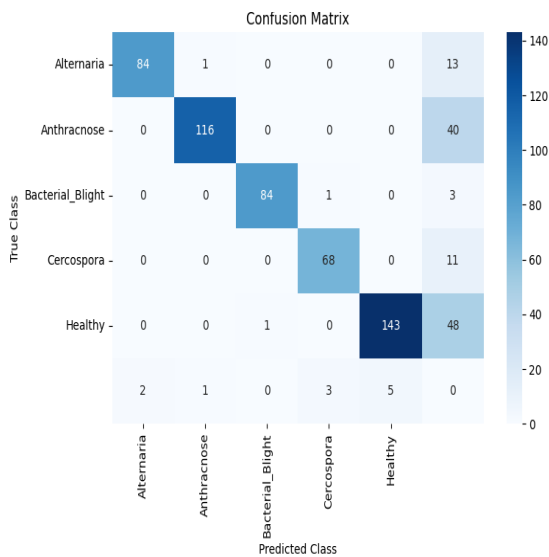


Figure 3. Confusion Matrix

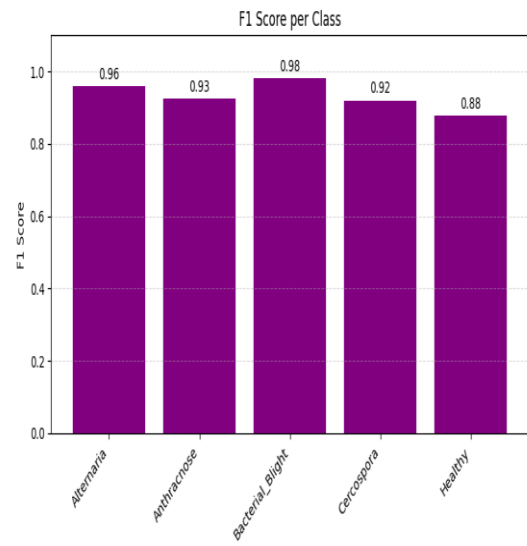


Figure 4. F1 Score per Class

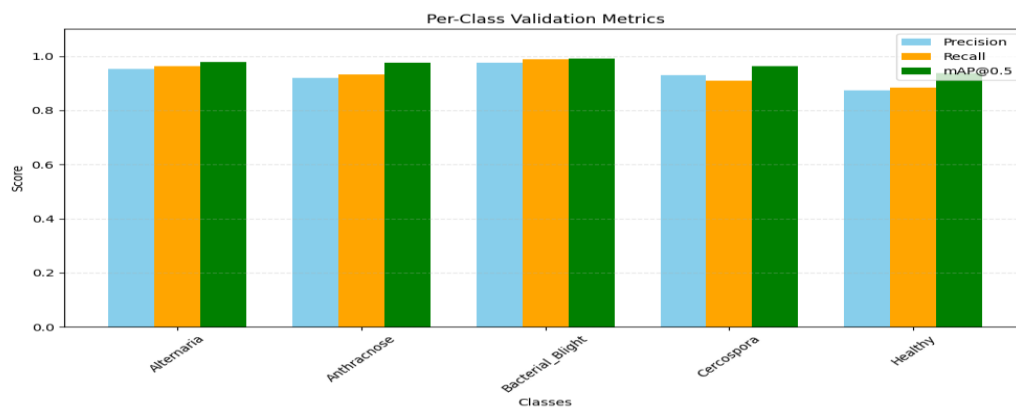


Figure 5. Per-Class Validation Metrics

7. Summary

Overall, the YOLOv12s model outperformed other variants and proved to be the most suitable for pomegranate disease detection. With its high accuracy, low inference time, and lightweight architecture, it provides a strong foundation for real-time agricultural monitoring systems. This model can be effectively deployed in smart farming applications using drones, mobile devices, or embedded systems, enabling early disease detection and reducing crop losses.

CONCLUSION

The project “Pomegranate Disease Detection Using YOLOv12: A Deep Learning Approach Across Five Classes” successfully demonstrates the application of advanced deep learning techniques for automated and accurate disease detection in pomegranate fruits. The study evaluated five YOLOv12 model variants — n, s, m, l, and x — using a standardized dataset containing 5,099 annotated images of five disease classes: *Healthy*, *Bacterial Blight*, *Anthrachnose*, *Cercospora Fruit Spot*, and *Alternaria Fruit Spot*.

From the comparative analysis, it was observed that the YOLOv12s variant achieved the highest overall performance with a Precision of 0.9300, Recall of 0.9357, and mAP@0.5:0.95 of 0.8620. It provided the best balance between accuracy, speed, and computational efficiency. While larger models like YOLOv12x and YOLOv12l offered comparable accuracy, they required more processing power and longer inference times, making them less suitable for real-time applications.

The results confirm that YOLOv12s is the most optimal variant for real-time and edge-based agricultural systems, capable of detecting multiple fruit diseases efficiently with minimal resource consumption. The model demonstrated consistent detection performance across all five pomegranate disease categories, showing high robustness even in challenging visual conditions such as lighting variations and occlusions.

This project highlights how artificial intelligence and deep learning can be integrated into precision agriculture to automate disease monitoring, reduce manual labor, and enable early intervention. The successful implementation of YOLOv12 in this context opens new possibilities for smart farming applications, including deployment on drones, smartphones, and embedded devices for field-level disease surveillance.

In conclusion, the system developed in this project provides a scalable, accurate, and efficient solution for fruit disease detection. It can be extended in future work by incorporating additional fruit types, expanding datasets, integrating IoT-based monitoring, or developing mobile applications for real-time disease reporting. This work contributes to the advancement of AI-driven agricultural technologies, promoting productivity, sustainability, and economic efficiency in modern farming.

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